

The Helix–Hilbert–Polya Bridge: Source Completeness and Projection Transport

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Abstract

This note records the mathematical core currently formalized in Lean and evaluated by the numerical model. The Hilbert–Polya side is the R1–R9 infrastructure chain. For non-principal channels it supplies source completeness: every nontrivial zero is realized by a three-dimensional source fiber event carrying the $\pi/3$ coordinate translation. The remaining projection input is faithful transport: realized source events survive projection, lower dimensions introduce no false events, and the projected event preserves the unit-circle/Pythagorean readout. With that transport input the formal closure gives GRH for non-principal channels. The zeta/L1 channel is handled separately through the eta-regularized zeta bridge with the analogous transport input.

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1 Objects and Notation

Let χ be a Dirichlet character modulo q and write

$$L(s, \chi)$$

for Mathlib's Dirichlet L -function. For non-principal χ , define the strip zero set

$$\text{NTZ}(\chi) := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, L(\rho, \chi) = 0\}.$$

The per-character GRH predicate formalized in Lean is

$$\text{GRH}(\chi) := \forall \rho \in \text{NTZ}(\chi), \quad \text{Re } \rho = \frac{1}{2}.$$

The all non-principal channel closure predicate used in the current Lean theorem is

$$\text{GRH}_{\neq 1} := \forall q, \forall \chi \pmod{q}, \quad \chi \neq 1 \implies \text{GRH}(\chi).$$

The principal zeta/L1 channel is treated separately through eta regularization. For zeta, let

$$\text{NTZ}_\zeta := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, \zeta(\rho) = 0\}.$$

2 The $\pi/3$ Helix Geometry

2.1 Canonical Scale

The numerical model and the formal gauge use the canonical $\pi/3$ chart

$$A = \frac{\pi}{6}, \quad ds = \frac{\pi}{3}, \quad C = 2A ds = \left(\frac{\pi}{3}\right)^2.$$

The exact area-law placement is

$$R_n^2 = Cn, \quad R_n = \frac{\pi}{3}\sqrt{n}.$$

At the source level the data are the $\pi/3$ coordinates, the channel weights coming from residue counting, and the harmonic phase on the helix. The fiber is derived from that source. No one-dimensional continuation formula is used to create it. The standard symbols L , ζ , and Hurwitz tails below occur because parts of the derived fiber identify with those representations after the geometric fiber has already been placed. Thus the three-dimensional source fiber atom at height t , for $s = \frac{1}{2} + it$, is

$$a_n(s) = w(n) \frac{1}{R_n} \exp\{-it \text{Log}(R_n^2)\} = w(n)(Cn)^{-s}.$$

Lemma 2.1 (Area-law closed form). *For any channel weights $w(n)$ for which the corresponding Dirichlet series is defined,*

$$F_w(s) := \sum_{n \geq 1} w(n)(Cn)^{-s} = C^{-s} \sum_{n \geq 1} w(n)n^{-s}.$$

Proof. Use $R_n^2 = Cn$ and $s = \frac{1}{2} + it$:

$$R_n^{-1} e^{-it \operatorname{Log}(R_n^2)} = (R_n^2)^{-1/2} (R_n^2)^{-it} = (R_n^2)^{-s} = (Cn)^{-s}.$$

Summing gives the result. □

2.2 Non-principal Character Channels

For a non-principal Dirichlet character χ modulo q , the channel weights are

$$w(n) = \chi(n).$$

Therefore the source channel sum is

$$F_\chi(s) = \sum_{n \geq 1} \chi(n)(Cn)^{-s}.$$

In the projected one-dimensional notation this same harmonic sum is

$$F_\chi(s) = C^{-s} L(s, \chi).$$

Residue-wise, with zero residues omitted, the same placed channel may be reindexed as

$$F_\chi(s) = C^{-s} q^{-s} \sum_{r \bmod q} \chi(r) \zeta\left(s, \frac{r}{q}\right).$$

For χ_3 ,

$$\chi_3(1) = 1, \quad \chi_3(2) = -1, \quad \chi_3(0) = 0,$$

and hence

$$F_{\chi_3}(s) = C^{-s} 3^{-s} \left[\zeta\left(s, \frac{1}{3}\right) - \zeta\left(s, \frac{2}{3}\right) \right].$$

2.3 The Principal/Zeta Channel

For the zeta/L1 channel the fiber is eta-regularized:

$$w(n) = (-1)^{n+1}.$$

The source channel sum is

$$F_{L1}(s) = \sum_{n \geq 1} (-1)^{n+1} (Cn)^{-s}.$$

In the projected one-dimensional notation the eta factor is

$$\eta(s) = \sum_{n \geq 1} (-1)^{n+1} n^{-s} = (1 - 2^{1-s}) \zeta(s).$$

Thus the L1 fiber is

$$F_{L1}(s) = C^{-s}\eta(s) = C^{-s}(1 - 2^{1-s})\zeta(s).$$

Equivalently, in residue form modulo 2,

$$F_{L1}(s) = C^{-s}2^{-s} \left[\zeta\left(s, \frac{1}{2}\right) - \zeta(s, 1) \right].$$

On the critical line $\text{Re } s = \frac{1}{2}$, the eta factor is nonzero:

$$1 - 2^{1-s} \neq 0,$$

because $2^{1-s} = 1$ would imply $2^{1-\text{Re } s} = 1$, hence $\text{Re } s = 1$.

3 Closed-Form Accumulation and Resolvent Trace

The placed fiber accumulation and the resolvent-trace accumulation are distinct but linked closed forms.

For a non-principal channel, the placed source fiber is

$$F_{\chi}^C(s) = \sum_{n \geq 1} \chi(n)(Cn)^{-s} = C^{-s}L(s, \chi).$$

This is the channel amplitude accumulation: it is the harmonic sum of the source fiber terms in the $\pi/3$ area chart.

Lemma 3.1 (No radial drift in the accumulation normal form). *The helix source has no radial drift by construction. For each source atom*

$$a_n(t) = w(n)R_n^{-1} \exp\{-it \text{Log}(R_n^2)\},$$

the radial size is independent of t :

$$|a_n(t)| = \frac{|w(n)|}{R_n} = \frac{|w(n)|}{(\pi/3)\sqrt{n}} \quad (C = (\pi/3)^2).$$

Proof. For real t and positive R_n^2 ,

$$|\exp\{-it \text{Log}(R_n^2)\}| = 1.$$

The helix flow therefore changes only phase. The radial coordinate is fixed by the source placement $R_n^2 = Cn$, so the accumulated source is no-drift in the radial direction before projection. $\square$

This no-drift normal form is the invariant used by the Polya induction. Moving from one crossing to the next advances the phase ledger and harmonic threshold, not the radial placement. Hence R7 can be applied to the infinite ladder with the same $\pi/3$ source geometry at every step; combined with R1–R6 and R8–R9, this is the source-side path to **SourceComplete3D**.

The resolvent trace is the logarithmic, prime-power-weighted accumulation. In the convergent half-plane,

$$\begin{aligned} \mathcal{T}_{\chi}^C(s) &:= \sum_{n \geq 1} \Lambda(n)\chi(n)(Cn)^{-s} \\ &= C^{-s} \sum_{n \geq 1} \Lambda(n)\chi(n)n^{-s} \\ &= C^{-s} \left(-\frac{L'}{L}(s, \chi) \right). \end{aligned}$$

Here Λ is the von Mangoldt weight. Thus the same source geometry has two closed-form accumulations: the fiber sum $C^{-s}L(s, \chi)$ and the resolvent trace $C^{-s}(-L'/L)(s, \chi)$. In Lean these are

$$\text{HelixGauge.helixSource_eq_gauge_mul_L}$$

and

$$\text{HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv.}$$

At the canonical scale $C = (\pi/3)^2$, the continued trace is

$$\mathcal{T}_\chi^{\pi/3}(s) = \left(\left(\frac{\pi}{3} \right)^2 \right)^{-s} \left(-\frac{L'}{L}(s, \chi) \right),$$

which is the paper form of `HelixGauge.piThirdHelixTraceCont_eq`.

The literal logarithmic derivative of the placed fiber differs only by a regular gauge term:

$$-\frac{d}{ds} \log F_\chi^C(s) = \log C - \frac{L'}{L}(s, \chi).$$

The additional term $\log C$ has no pole. The multiplicative geometric gauge C^{-s} is never zero for $C > 0$. Therefore neither gauge changes the crossing locations or creates/removes trace poles; the multiplicative gauge only rescales the trace residue by a nonzero geometric factor.

This gives the local residue law at a capture event. If ρ is a zero of $L(s, \chi)$ of order m and

$$L(s, \chi) = (s - \rho)^m g(s), \quad g(\rho) \neq 0,$$

then near ρ

$$-\frac{L'}{L}(s, \chi) = -\frac{m}{s - \rho} - \frac{g'}{g}(s),$$

and hence

$$\mathcal{T}_\chi^C(s) = -\frac{mC^{-\rho}}{s - \rho} + \text{a term finite at } \rho.$$

So the trace pole has residue $-mC^{-\rho}$. The integer m is the spectral multiplicity; the factor $C^{-\rho}$ is the faithful $\pi/3$ geometric scaling. At a crossing, this is the residue consumed by the resolvent trace while the source fiber amplitude crests, the channel phase cancels, and the sign changes. The Polya threshold condition is therefore the simultaneous event

$$\begin{aligned} F_\chi^C(\rho) &= 0, & \text{Res}_{s=\rho} \mathcal{T}_\chi^C(s) &= -mC^{-\rho}, \\ E_\chi(\text{Im } \rho) &= A_k^\pm, & A_0^\pm &= \pm \frac{\pi}{2}, & A_{k+1}^\pm - A_k^\pm &= \pm \pi. \end{aligned}$$

Here E_χ is the source accumulation ledger used by the Polya chain. The jump from one level $A_k^\pm$ to the next is the threshold that starts the next harmonic on the spectral readout. Thus the closed-form resolvent residue, the crossing sign change, and the creation of the next harmonic are the same source event viewed in three ledgers: trace, standing wave, and harmonic ladder.

For primitive non-principal χ , the completed zero-side resolvent trace is the partial-fraction accumulation

$$\mathcal{R}_\chi(s) = \sum_{\rho \in \text{NTZ}(\chi)} m_\rho \left(\frac{1}{s - \rho} + \frac{1}{\rho} \right),$$

where m_ρ is the vanishing order of the completed L -function at ρ . The formal closed form is

$$\log \text{Deriv } \Lambda_\chi(s) = A + \mathcal{R}_\chi(s) \quad (s \notin \text{NTZ}(\chi)),$$

for a constant A . Splitting the completed function $\Lambda_\chi = \gamma_\chi L(s, \chi)$ gives, off the zeros in the strip,

$$-\frac{L'}{L}(s, \chi) = -A - \mathcal{R}_\chi(s) + \log \text{Deriv } \gamma_\chi(s).$$

Thus the prime-power accumulation, the zero-side resolvent trace, and the continued source trace are the same readout up to the constant and Archimedean gauge terms, which are regular away from their own gauge poles.

For the L1/zeta channel,

$$F_{L1}^C(s) = C^{-s}(1 - 2^{1-s})\zeta(s).$$

Off $\text{Re } s = 1$, the eta factor is nonzero, so a zero of the L1 fiber is a zero of ζ . The corresponding resolvent trace is

$$-\frac{\zeta'}{\zeta}(s),$$

and the $\pi/3$ zeta fiber zero has a resolvent-trace pole by

$$\text{HelixGauge.piThirdZetaFiber_zero_resolventTrace_pole.}$$

4 Standing Readout and Projection to One Dimension

This is the layer where the placed three-dimensional fiber is read in the lower-dimensional coordinate chart. The functions below are readouts of the source geometry and channel counts, with the coordinate phase correction kept explicit.

For $s = \frac{1}{2} + it$, the projected complex fiber is

$$F_\chi(t) = F_\chi\left(\frac{1}{2} + it\right).$$

The real standing readout has the form

$$Z_\chi(t) = \text{Re} \left(e^{i\Theta_\chi(t)} F_\chi\left(\frac{1}{2} + it\right) \right),$$

where

$$\Theta_\chi(t) = \arg \Gamma\left(\frac{1/2 + a}{2} + \frac{it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi} + t \log C + \alpha_\chi.$$

Here $a \in \{0, 1\}$ is the parity and α_χ is the character/root-number gauge. For the L1/zeta eta-regularized channel one subtracts the eta gauge

$$\arg(1 - 2^{1-s}).$$

The term $t \log C$ is not a new oscillation. It is the coordinate correction from

$$\text{Log}(R_n^2) = \text{Log}(Cn) = \log n + \log C.$$

Omitting it rotates the real readout into the wrong coordinate chart.

4.1 Phase Conventions

All logarithms of positive real quantities are real logarithms. Thus

$$\text{Log}(R_n^2) = \log(Cn), \quad \log q, \quad \log C$$

use the real branch. The gamma phase in the standing readout is the continuous phase

$$\vartheta_\chi(t) = \text{Im} \log \Gamma\left(\frac{1/2 + a + it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi},$$

with the branch chosen continuously along the vertical line $z(t) = (1/2 + a + it)/2$. The root-number phase is fixed by

$$\tau(\chi) = \sum_{r=1}^q \chi(r) e^{2\pi i r/q}, \quad \varepsilon_\chi = \frac{\tau(\chi)}{i^a \sqrt{q}}, \quad \alpha_\chi = -\frac{1}{2} \arg(\varepsilon_\chi).$$

For the principal zeta/L1 channel, $\alpha_\chi = 0$ and the two-channel eta readout uses the continuous phase

$$\phi_\eta(t) = \arg\left(1 - 2^{1-(1/2+it)}\right),$$

so that

$$\Theta_{L1}(t) = \vartheta_\zeta(t) + t \log C - \phi_\eta(t).$$

For non-principal χ ,

$$\Theta_\chi(t) = \vartheta_\chi(t) + t \log C + \alpha_\chi.$$

5 Harmonic Cost and the Quantum Ladder

The full source geometry is double ended. The midpoint is the origin of the fiber, and the reflected side is an anti-helix carrying the opposite orientation. The proof uses the positive oriented side of the helix for simplicity; the reflected anti-helix is not used in the formal proof closure. It is used in computational verification: the reflected side supplies the symmetric pre-origin geometry needed to build a high-precision tail before the counting ledger starts at the origin. Its cost law is the same with the sign reversed.

The crossing levels in the projected unit-circle readout are therefore

$$A_k = \left(k + \frac{1}{2}\right) \pi, \quad k \in \mathbb{Z}.$$

Equivalently, for $m \geq 0$ the two oriented ladders are

$$A_m^+ = \left(m + \frac{1}{2}\right) \pi, \quad A_m^- = -\left(m + \frac{1}{2}\right) \pi.$$

On the positive helix side used in the proof,

$$A_0 = \frac{\pi}{2}, \quad A_{k+1} = A_k + \pi.$$

Thus the first crossing is the only crossing on that side whose cost from the origin is a half quantum:

$$|A_0^+| = \frac{\pi}{2}.$$

Every additional crossing on that same side costs one full quantum:

$$A_{m+1}^+ - A_m^+ = \pi \quad (m \geq 0).$$

The reflected anti-helix obeys the same rule:

$$|A_0^-| = \frac{\pi}{2}, \quad |A_{m+1}^- - A_m^-| = \pi \quad (m \geq 0).$$

So the half-quantum occurs once at the first crossing on each oriented side of the double-ended helix; all subsequent crossings consume full π increments.

Conjecture 5.1 (Origin/Tate interface). *If this helix model has a Tate-thesis-style global/local interpretation, the natural place for that connection is the shared origin of the double-ended helix. The origin is where the two reflected orientations meet before the one-dimensional readout separates them into positive and negative crossing ladders.*

6 Hilbert–Polya Infrastructure: R1–R9

This section lists the infrastructure available before the projection-faithfulness hypothesis is supplied. It does not assert GRH.

R1. Hilbert space. The source space is

$$\ell^2(\mathbb{N}; \mathbb{C}).$$

R2. Unitary phasor flow. The FTA/Euler phasor flow is

$$U(t)(n) = n^{it}, \quad |U(t)(n)| = 1,$$

with

$$U(s+t)(n) = U(s)(n)U(t)(n).$$

R3. Real generator and no-drift principle. The generator is

$$H(n) = \log n \in \mathbb{R}, \quad U(t)(n) = e^{itH(n)}.$$

The source no-drift lemma is: if

$$e^{\operatorname{Re} \lambda \tau} c = c \quad \text{for all } \tau, \quad c \neq 0,$$

then

$$\operatorname{Re} \lambda = 0.$$

R4. Helix-native trace/readout equals L . The finite phasor chain is

$$S_M(s, \chi) = \sum_{n \leq M} \chi(n) n^{-\operatorname{Re} s} U(\operatorname{Im} s)(n)^{-1}.$$

For $0 < \operatorname{Re} s$ and $\chi \neq 1$,

$$S_M(s, \chi) \rightarrow L(s, \chi)$$

with explicit rate:

$$\|S_M(s, \chi) - L(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

R5. **Zeros induce fiber capture events.** At a zero $L(s, \chi) = 0$ with $0 < \operatorname{Re} s$,

$$\|S_M(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

At every $\rho \in \operatorname{NTZ}(\chi)$,

$$-\frac{L'}{L}(s, \chi)$$

has no finite limit at ρ in the punctured neighborhood.

R6. **Spectral realization by crossing confluence and channel cancellation.** For any continuous strictly monotone accumulation E , if

$$E(a) \leq c \leq E(b), \quad a \leq b,$$

then there is a unique t such that

$$a \leq t, \quad E(t) = c.$$

At threshold $E(t) = n\pi$,

$$\left\lfloor \frac{E(t)}{\pi} \right\rfloor = n.$$

At a crossing threshold the source event is a confluence: the channel phase cancels, the amplitude reaches its crest, the resolvent trace consumes the local residue, the sign changes, and the next harmonic begins on the spectral readout. In the source normalization used by the Polya chain,

$$E(\operatorname{Im} \rho) = A_k^\pm, \quad A_0^\pm = \pm \frac{\pi}{2}, \quad A_{k+1}^\pm - A_k^\pm = \pm \pi.$$

So the first crossing on each oriented side costs a half quantum, and every subsequent threshold is one full π step. The channel cancellation law is:

$$\sum_{m \in F} s_m = 0 \iff P_F(s) = M_F(s),$$

where

$$P_F(s) = \sum_{m \in F} \max(s_m, 0), \quad M_F(s) = \sum_{m \in F} \max(-s_m, 0).$$

At cancellation,

$$\sum_{m \in F} |s_m| = 2P_F(s).$$

R7. **Exhaustion by ladder induction and rigidity.** If a property holds at A_0 and is preserved by $A_k \mapsto A_{k+1}$ and $A_k \mapsto A_{k-1}$, then it holds at every A_k . Moreover, any crossing sequence c_n satisfying

$$E(c_n) = n\pi$$

is forced to be the purchase-height sequence:

$$c_n = \operatorname{purchaseHeight}(n).$$

R8. Coordinate faithfulness and Pythagorean readout. The rechart

$$\text{arcChart}(x) = \frac{\pi}{3}x$$

has inverse

$$\text{arcChartInv}(u) = \frac{3}{\pi}u.$$

The critical midpoint is

$$\text{arcChart}(x) = \frac{\pi}{6} \iff x = \frac{1}{2}.$$

The value $1/2$ is therefore only the normalized one-dimensional readout of the source midpoint $\pi/6$. It is not an independently fitted spectral constant. At a source crossing, the standing readout changes sign; that sign change forces the crossing to occur at the midpoint of the source chart. The rule is source-side: projection is only responsible for transporting the event without losing it or introducing a new one. The midpoint conclusion routes through the unitary/Pythagorean readout. In natural coordinates the readout satisfies

$$(\text{readout}_{\sigma,\gamma})_{\text{Re}}^2 + (\text{readout}_{\sigma,\gamma})_{\text{Im}}^2 = 1 \iff \sigma = \frac{1}{2}.$$

Equivalently, for the spectral value $w(\rho) = 1 - 1/\rho$,

$$\|w(\rho)\|^2 = 1 \iff \text{Re } \rho = \frac{1}{2}.$$

The source conservation/no-drift input supplies the unitary source side of this readout.

R9. Genuine zero-set object. The object is Mathlib's L -function zero set:

$$\rho \in \text{NTZ}(\chi) \iff 0 < \text{Re } \rho < 1, \quad L(\rho, \chi) = 0.$$

7 Source Completeness and Projection Faithfulness

Definition 7.1 (Source crossing confluence). *For a channel χ and point $\rho = \sigma + i\gamma$, a source crossing confluence at height γ is a three-dimensional fiber event with the following simultaneous features:*

- (C1) *the channel phases cancel, so the fiber vanishes in the channel sum;*
- (C2) *the accumulated amplitude reaches a crest at one of the ladder levels $A_m^\pm$;*
- (C3) *the crossing consumes the threshold energy required to start the next harmonic on the spectral readout;*
- (C4) *the real standing readout changes sign at the crossing;*
- (C5) *the source midpoint coordinate forced by the sign change and the real-axis/unitary readout requirement is $x = \pi/6$ in the $\pi/3$ chart, while the height γ is retained as the vertical coordinate.*

Thus the source crossing itself determines the midpoint. Clause (C5) records that consequence; it is not an additional lower-dimensional placement assumption. After the normalization $x \mapsto 3x/\pi$, this source midpoint is the value $1/2$.

Under projection from the three-dimensional fiber to the unit-circle readout, radial and absolute phase information are collapsed. The retained datum is the crossing height γ , together with the unit-circle event. The subsequent one-dimensional readout rescales the midpoint coordinate by

$$x \mapsto \frac{3}{\pi}x,$$

so that the source midpoint $x = \pi/6$ is read as $1/2$.

Definition 7.2 (Fiber capture event). *For a Dirichlet channel χ ,*

$$\text{FiberCaptureEvent}_\chi(\rho) \quad :\Longleftrightarrow \quad \rho \in \text{NTZ}(\chi).$$

Definition 7.3 (Source fiber event). *For a non-principal Dirichlet channel χ and point ρ , a source fiber event is a three-dimensional event over ρ carrying the following data:*

(S1) *a source drift rate λ_ρ and nonzero amplitude a_ρ with*

$$e^{\text{Re}(\lambda_\rho)\tau} a_\rho = a_\rho \quad (\forall \tau \in \mathbb{R});$$

(S2) *the $\pi/3$ source coordinate*

$$x_\rho = \text{arcChart}(\text{Re } \rho) = \frac{\pi}{3} \text{Re } \rho;$$

(S3) *height retention*

$$\text{Im}(\lambda_\rho) = \text{Im } \rho;$$

(S4) *nonzero spectral value $\rho \neq 0$;*

(S5) *source-chain cancellation with explicit decay;*

(S6) *the resolvent-trace pole at ρ .*

It does not contain the conclusion $\text{Re } \rho = 1/2$.

Theorem 7.4 (Source completeness from R1–R9). *For every non-principal χ , every fiber capture event $\rho \in \text{NTZ}(\chi)$ is realized by a source fiber event.*

Proof. This is the Lean theorem `sourceComplete3D_of_HP`. It uses: R5 for source-chain cancellation and the resolvent pole, and R9 to identify the Mathlib zero-set object. The produced source event records the R8 coordinate language by setting its source coordinate to $\text{arcChart}(\text{Re } \rho) = (\pi/3) \text{Re } \rho$. The midpoint translation $\text{arcChart}(\text{Re } \rho) = \pi/6 \iff \text{Re } \rho = 1/2$ is used later in `SourceFiberEvent.midpoint_axis_of_HP`. Source completeness itself does not prove $\text{Re } \rho = 1/2$. $\square$

Hypothesis 7.5 (Faithful dimensional projection for one channel). *There is a predicate $\text{Proj}_\chi(\rho)$ of projected crossing events such that:*

(F1) *Every realized source fiber event is faithfully projected:*

$$\text{SourceFiberEvent}_\chi(\rho) \implies \text{Proj}_\chi(\rho).$$

(F2) *Lower-dimensional projection introduces no false crossing events:*

$$\text{Proj}_\chi(\rho) \implies \rho \text{ is represented by a source fiber event.}$$

(F3) *The projected event preserves the unit-circle/Pythagorean readout:*

$$\text{Proj}_\chi(\rho) \implies \|w(\rho)\|^2 = 1, \quad w(\rho) = 1 - \frac{1}{\rho}.$$

Thus the projection input is only faithfulness of dimensional collapse at crossings: realized source events survive transport to the lower-dimensional readout, no false events are introduced, and the transported event preserves the unit-circle readout. Source completeness itself is supplied by R1–R9.

8 Closure Under Faithful Transport

Theorem 8.1 (Per-character closure). *Let $\chi \neq 1$. Assume faithful dimensional projection for χ . Then*

$$\text{GRH}(\chi).$$

Proof. Let $\rho \in \text{NTZ}(\chi)$. By source completeness from R1–R9, ρ is realized by a source fiber event. R3 gives source no-drift from the conserved amplitude. Faithful projection transports that source event to the one-dimensional readout and preserves

$$\|w(\rho)\|^2 = 1.$$

The Lean theorem `SpectralSide.w.unit_iff_half` gives

$$\text{Re } \rho = \frac{1}{2}.$$

The numerical value $1/2$ enters here only because the source crossing has forced the midpoint and R8 reads the $\pi/3$ midpoint $\pi/6$ in unit coordinates. The projection hypothesis does not create this value; it only transports the source event and preserves the unit-circle readout. Equivalently, R8 records the $\pi/3$ chart identity

$$\text{arcChart}(\text{Re } \rho) = \frac{\pi}{6} \iff \text{Re } \rho = \frac{1}{2}.$$

Since ρ was arbitrary, $\text{GRH}(\chi)$ follows. □

Theorem 8.2 (All non-principal channel closure). *Assume faithful dimensional projection for every non-principal Dirichlet channel. Then every non-principal channel satisfies*

$$\text{GRH}(\chi).$$

Proof. Apply the per-character closure to each modulus and each $\chi \neq 1$. □

Theorem 8.3 (Zeta/L1 transport closure). *Assume the corresponding eta-regularized zeta/L1 projection transport. Then Mathlib’s `RiemannHypothesis` holds.*

Proof. For $\rho \in \text{NTZ}_\zeta$, the zeta projection-faithfulness hypothesis gives a source fiber event. It is registered at $x = \pi/6$ in the $\pi/3$ chart, and the normalized Pythagorean readout is

$$\text{Re } \rho = \frac{1}{2}.$$

The existing zeta bridge converts

$$\forall \rho \in \text{NTZ}_\zeta, \quad \text{Re } \rho = \frac{1}{2}$$

to Mathlib’s `RiemannHypothesis`. □

9 Mathematical Reproduction of the Computation

The computation evaluates the source fiber equations above. It is not a proof step. The crossing ordinates are defined by the complete geometric fiber and its standing readout. Finite summation, residue decomposition, Hurwitz notation, and Euler–Maclaurin summation appear only where the derived fiber is represented for one-dimensional evaluation; they are not sources of the fiber.

9.1 Placed Geometric Fiber

The geometry is

$$A = \pi/6, \quad ds = \pi/3, \quad C = 2A ds = (\pi/3)^2.$$

For $s = \frac{1}{2} + it$, the placed finite head is

$$H_M(t) = \sum_{n=1}^M \chi(n) R_n^{-1} e^{-it \operatorname{Log}(R_n^2)}, \quad R_n^2 = Cn.$$

Thus

$$H_M(t) = \sum_{n=1}^M \chi(n) (Cn)^{-s}.$$

The tail is not added from a separate external object. It is the remaining geometric fiber beyond the cutoff:

$$G_M(t) = \sum_{n>M} \chi(n) R_n^{-1} e^{-it \operatorname{Log}(R_n^2)} = \sum_{n>M} \chi(n) (Cn)^{-s}.$$

The complete placed fiber is

$$F_\chi(t) = H_M(t) + G_M(t),$$

independently of the cutoff M when the tail is included.

9.2 Residue Coordinates for the Geometric Tail

The geometric tail may be written in conductor residue coordinates. Let $n_{0,r}$ be the first integer $> M$ in residue class $r \bmod q$. Then

$$G_M(t) = \sum_r \chi(r) C^{-s} q^{-s} \sum_{m \geq 0} \left(\frac{n_{0,r}}{q} + m \right)^{-s}.$$

This identity is just the residue-class reindexing of the geometric terms $R_n^2 = Cn$. Hurwitz notation abbreviates the inner residue tail; it does not define the tail or the crossing condition. It is a coincidental representation of the derived fiber.

For finite one-dimensional evaluation, replace the inner infinite residue tail by

$$\mathcal{T}_{h,J}(s, x) \approx \sum_{m \geq 0} (x + m)^{-s}.$$

For integers $h, J \geq 1$, set $X = x + h$ and write

$$(s)^{\overline{m}} = s(s+1) \cdots (s+m-1), \quad (s)^{\overline{0}} = 1.$$

The Euler–Maclaurin summation evaluator used for that geometric residue tail is

$$\begin{aligned}\mathcal{T}_{h,J}(s, x) &= \sum_{m=0}^{h-1} (x+m)^{-s} + \frac{X^{1-s}}{s-1} + \frac{1}{2}X^{-s} \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{(2j)!}(s)^{\overline{2j-1}}X^{-s-2j+1},\end{aligned}$$

where B_{2j} are Bernoulli numbers. The associated remainder is the standard Euler–Maclaurin remainder after the B_{2J} term. The numerical tables in the repository use $h = 8$ and finite J chosen large enough for the working precision.

With this evaluator, the finite computed fiber is

$$F_{M,\chi}(t) = \sum_{n=1}^M \chi(n)(Cn)^{-s} + \sum_r \chi(r)C^{-s}q^{-s}\mathcal{T}_{h,J}\left(s, \frac{n_{0,r}}{q}\right).$$

For the L1 channel, the same geometric tail is grouped with conductor 2 and weights

$$\chi_\eta(1) = 1, \quad \chi_\eta(0) = -1,$$

so that

$$F_{M,L1}(t) = C^{-s}2^{-s} \left[\mathcal{T}_{h,J}\left(s, \frac{n_{0,1}}{2}\right) - \mathcal{T}_{h,J}\left(s, \frac{n_{0,0}}{2}\right) \right] + \sum_{n=1}^M (-1)^{n+1}(Cn)^{-s}.$$

This is the finite evaluation of the eta-regularized zeta fiber in placed geometric coordinates.

9.3 Principal One-Dimensional Tail Error Control

For the projected principal/L1 tail, the repository also formalizes the classical Euler–Maclaurin summation identity as an error-control theorem. Let

$$S_N(s) = \sum_{n=1}^N n^{-s}, \quad \{u\} = u - \lfloor u \rfloor,$$

and define

$$I(s, N) = \int_N^\infty \{u\} u^{-s-1} du, \quad R(s, N) = sI(s, N).$$

For $\operatorname{Re} s > 0$ and $s \neq 1$, the continued one-dimensional response is

$$c(s) = \frac{s}{s-1} - s \int_1^\infty \{u\} u^{-s-1} du,$$

and the Euler–Maclaurin identity is

$$c(s) = \zeta(s),$$

with finite-readout error

$$S_N(s) - \zeta(s) - \frac{N^{1-s}}{1-s} = R(s, N)$$

and bound

$$\|R(s, N)\| \leq \frac{\|s\|}{\operatorname{Re} s} N^{-\operatorname{Re} s} \quad (N \geq 2).$$

This theorem controls the finite projected evaluation of the principal geometric tail; it is a representation of the derived fiber, not a source for the tail.

The full finite readout is

$$Z_\chi(t) = \operatorname{Re}(e^{i\Theta_\chi(t)} F_{M,\chi}(t)).$$

Crossings are isolated by sign changes

$$Z_\chi(a)Z_\chi(b) < 0$$

followed by bisection. If $\gamma \in (a, b)$ is a simple crossing,

$$Z_\chi(\gamma) = 0, \quad Z'_\chi(\gamma) \neq 0,$$

then there is an interval around γ on which the sign changes once, and bisection on any bracket inside that interval converges to γ .

For local reproduction from an intermediate height t_0 , choose $\delta > 0$ and evaluate Z_χ on $[t_0 - \delta, t_0 + \delta]$. If a sign change occurs in $[a, b] \subset [t_0 - \delta, t_0 + \delta]$, bisection on Z_χ rederives the crossing in that local interval; no earlier crossing is used.

10 Formal and Computational Support

The proof above is the mathematical statement. The repository support has two roles: the formal files check the stated implications in Lean, and the numerical files evaluate the displayed fiber equations.

10.1 Formalization Map

Dirichlet zero set	GRHSpectral.NontrivialZeros
per-character GRH predicate	GRHSpectral.GRH
zeta zero set	ZD.NontrivialZeros
Euler–Maclaurin continuation	GRH.EulerMaclaurinDirichlet.c_fun
Euler–Maclaurin error term	GRH.EulerMaclaurinDirichlet.R
Euler–Maclaurin identity	GRH.EulerMaclaurinDirichlet.difference_eq_R
Euler–Maclaurin bound	GRH.EulerMaclaurinDirichlet. euler_maclaurin_dirichlet_explicit
zeta one-dimensional completion	SourceContinuation.zetaCompletion
residue fiber decomposition	DirichletFibre.LFunction_eq_fibre_sum
fiber cancellation	DirichletFibre.zero_iff_fibre_cancellation
source no-drift	HelixSource.source_noDrift
prime flow trace	HelixFlowVonMangoldt.flowVonMangoldtTrace_eq_ negLogDeriv
Möbius unit readout	moebius_unit_iff
Pythagorean readout	readout_pythagoras_iff_online
spectral unit midpoint	SpectralSide.w_unit_iff_half
$\pi/3$ gauge	HelixGauge.piThirdGauge_eq
nonprincipal closed form	HelixGauge.piThirdHelixSource_eq_gauge_mul_L
geometric trace closed form	HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv
$\pi/3$ continued trace	HelixGauge.piThirdHelixTraceCont_eq
completed trace gauge split	completedLFunction_logDeriv_gauge_split
prime/dual trace meeting	flowDualTraceMeeting
strip trace continuation	primeTraceContinuationIntoStrip
L1 closed form	HelixGauge.piThirdZetaFiber
L1 zero equivalence	HelixGauge.piThirdZetaFiber_zero_iff
L1 resolvent pole	HelixGauge.piThirdZetaFiber_zero_resolventTrace_ pole
R1–R9 infrastructure	hilbertPolyaChainR1ToR9
source fiber event	SourceFiberEvent
source completeness predicate	SourceComplete3D
source completeness from R1–R9	sourceComplete3D_of_HP
projection faithfulness	FaithfulDimensionalProjection
source midpoint from HP and transport	SourceFiberEvent.midpoint_axis_of_HP
per-character closure	GRH_of_HP_and_Faithfulness
all non-principal channel closure	GRHComplete_of_HP_and_Faithfulness
zeta/L1 closure	RH_of_HP_and_Faithfulness

10.2 Lean Verification Protocol

File-level verification:

```
lake env lean RequestProject/HilbertPolyaChain.lean
```

Closed-form gauge verification:

```
lake env lean RequestProject/HelixGaugeIdentity.lean
```

Escape-hatch audit:

```
rg -n "axiom |sorry|admit|@[implemented_by\]" \  
RequestProject/HilbertPolyaChain.lean
```


Axiom footprint targets:

```
#print axioms hilbertPolyaChainR1ToR9,
#print axioms HelixFlowVonMangoldt.flowVonMangoldtTrace_eq_negLogDeriv,
#print axioms HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv,
#print axioms completedLFunction_logDeriv_gauge_split,
#print axioms flowDualTraceMeeting,
#print axioms primeTraceContinuationIntoStrip,
#print axioms sourceComplete3D_of_HP,
#print axioms SourceFiberEvent.midpoint_axis_of_HP,
#print axioms GRH_of_HP_and_Faithfulness,
#print axioms GRHComplete_of_HP_and_Faithfulness,
#print axioms ZetaSourceFiberEvent.midpoint_axis_of_source_noDrift,
#print axioms RH_of_HP_and_Faithfulness,
#print axioms HelixGauge.piThirdGauge_eq,
#print axioms HelixGauge.piThirdZetaFiber_zero_iff,
#print axioms HelixGauge.piThirdZetaFiber_zero_resolventTrace_pole.
```

Expected footprint:

[propext, Classical.choice, Quot.sound].

11 Current Logical Status

HP R1–R9 infrastructure	Lean theorem	hilbertPolyaChainR1ToR9
$\pi/3$ closed form	Lean theorems	HelixGauge.piThird*
Source completeness	Lean theorem	sourceComplete3D_of_HP
Projection transport atoms	structure fields	FaithfulDimensionalProjection
Midpoint from source and transport	Lean theorem	SourceFiberEvent.midpoint_axis_of_HP
Per-character non-principal closure	transport theorem	GRH_of_HP_and_Faithfulness
All non-principal channel closure	transport theorem	GRHComplete_of_HP_and_Faithfulness
Zeta/L1 closure	transport theorem	RH_of_HP_and_Faithfulness

The projection transport atoms are exactly the following fields:

source_event_faithfully_projected	$\text{SourceFiberEvent}_\chi(\rho) \Rightarrow \text{Proj}_\chi(\rho)$
lower_dimensions_create_no_events	$\text{Proj}_\chi(\rho) \Rightarrow \text{SourceFiberEvent}_\chi(\rho)$
projected_readout_pythagorean	$\text{Proj}_\chi(\rho) \Rightarrow \ w(\rho)\ ^2 = 1$

All other atoms used in the non-principal closure are supplied by the HP chain, the $\pi/3$ coordinate chart, the source no-drift lemma, or the Mathlib zero-set definitions listed above.